💡 Project Title:

MAITRI – AI Assistant for Psychological & Physical Well-Being

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🧭 Project Brief / Overview:

MAITRI is an intelligent web application designed to assist individuals in monitoring and improving their **mental and emotional well-being** through **AI-based emotion analysis**.

The system analyzes a user's **facial expressions** and **voice tone** to detect emotions such as *happy, sad, angry, fear, surprise,* or *neutral*. Based on the detected mood, the app provides personalized **mental health tips**, **motivational quotes**, or even **relaxation exercises (like breathing or meditation suggestions)**.

This project bridges the gap between **technology and mental health** — providing users an accessible, real-time emotion support system directly from their browser.

⚙️ Core Functionalities / Features:

🎭 1. Facial Emotion Recognition

- Captures the user's face using a webcam.
- Supports emotions like *happy, sad, angry, disgust, fear, neutral,* etc.
- Enhanced with **OpenCV face detection** for accurate face cropping and analysis.

🎙️ 2. Voice Emotion Detection *(next module / optional add-on)*

- Records short audio clips via microphone.
- Uses a speech-emotion recognition model to detect tone-based emotions.
- Helps identify stress or anxiety through vocal features (pitch, intensity, tone).

🧘 3. Personalized Well-Being Suggestions

- Based on detected emotion, MAITRI recommends:
  - Motivational quotes
  - Calming music or meditation playlists
  - Breathing or relaxation exercises

- Short activity suggestions (e.g., “Take a short walk”, “Hydrate yourself”)

📊 4. Emotion History Tracking

- Stores recent emotion results to visualize emotional trends over time.
- User can view a simple graph of mood fluctuations.

🌐 5. Responsive Web Interface

- Simple, interactive frontend built using HTML, CSS, and JavaScript.
- Real-time camera and microphone access through browser APIs.
- Displays prediction results instantly with confidence levels.

🔒 6. Privacy-Friendly

- All analysis is done locally on the user’s browser + Flask backend.
- No personal data stored or shared externally.

🧩 Workflow / Architecture:

1. User opens MAITRI web app.
2. Webcam captures the face (or mic captures audio).
3. Image/audio is sent to Flask backend.
4. Flask processes input → detects face → runs AI model → predicts emotion.
5. Emotion + confidence returned to frontend.
6. Frontend displays result & suggests suitable actions/tips.